S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR

Practical No. 07

Aim: Write a Prolog Program to demonstrate the concept of Family relationship.

Name of Student: _____

Roll No.: _____

Semester/Year: **VI/III**

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Date of Performance: _____

Date of Submission: _____

AIM: Write a Prolog Program to demonstrate the concept of Family relationship.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Represent knowledge using **facts**
- Define rules using **logical inference**
- Understand **unification**
- Observe **backtracking**
- Model **hierarchical relationships** (parent, sibling, grandparent, etc.)

THEORY:

Prolog (Programming in Logic) is a **declarative logic programming language** based on:

- First-Order Predicate Logic
- Resolution
- Unification
- Backtracking

Unlike procedural languages (C, Python), Prolog focuses on **what is true**, not **how to compute it**.
Common implementation: **SWI-Prolog** In
Prolog a program consists of:

1. Facts
2. Rules
3. Queries

Installing Prolog (Using SWI-Prolog)

Step 1: Download

Visit: https://www.swi-prolog.org

Step 2: Install

Follow OS-specific installer.

Step 3: Run

Open terminal / command prompt:

swipl

?- This is the Prolog query prompt.

Prolog data types:

- **Atoms** → `john`, `apple`
- **Variables** → `X`, `Person`
- **Numbers** → `10`, `3.14`
- **Structures** → `parent(john, mary)`
- **Lists** → `[1,2,3]` **Unification:**

Prolog matches patterns for unification. Exa:

parent (john, X) Unifies with: parent (john, mary)

Result:

```
X = mary
```

Backtracking:

Prolog automatically searches for alternatives using backtracking. Example Query:

?- parent (john, X).

Press `;` to get next solution.

Important Operators:

Operator	Meaning
`:-`	If
`,`	AND
`;`	OR
`=`	Unification
`is`	Arithmetic evaluation
`\=`	Not unifiable
`\+`	Negation as failure

A **family relationship system** is a classical example used in Artificial Intelligence to demonstrate:

- Knowledge representation
- Predicate logic
- Recursive reasoning
- Depth-first search with backtracking

ALGORITHM:

1. Start

2. Define family members using `male/1` and `female/1` facts 3. Define parent relationships
4. Define derived relationships using rules:

    a. father
    b. mother
    c. sibling
    d. grandparent
    e. ancestor

5. Load the program in Prolog
6. Execute queries
7. Observe results and backtracking
8. Stop

PROGRAM CODE:

```
% --------- Facts (Family Members) ---------

male(john).
male(paul).
male(mike).
male(david).

female(mary).
female(linda).
female(susan).
female(anna).

parent(john, paul).
parent(mary, paul).

parent(john, linda).
parent(mary, linda).

parent(paul, mike).
parent(susan, mike).

parent(paul, anna).
parent(susan, anna).

parent(linda, david).

% --------- Rules (Relationships) ---------

% Father
father(X, Y) :-
   parent(X, Y),
   male(X).

% Mother
mother(X, Y) :-
   parent(X, Y),
```

```
    female(X).

% Child
child(X, Y) :-
    parent(Y, X).

% Sibling
sibling(X, Y) :-
    parent(Z, X),
    parent(Z, Y),
    X \= Y.

% Brother
brother(X, Y) :-
    sibling(X, Y),
    male(X).

% Sister
sister(X, Y) :-
    sibling(X, Y),
    female(X).

% Grandparent
grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).

% Grandfather
grandfather(X, Y) :-
    grandparent(X, Y),
    male(X).

% Grandmother
grandmother(X, Y) :-
    grandparent(X, Y),
    female(X).
```

INPUT & OUTPUT:

Step 1: Start Prolog swipl
Step 2: Consult File
?- consult('family.pl'). OR

?- [family].

Queries & Output
1. Find father of Sam ?-
father(X, sam).
Output:
X = john.

2. Find siblings of Sam ?-
sibling(sam, X).
Output:
X = anna.

3. Find grandparents of Paul ?-

grandparent(X, paul).

Output:

X = john ; X =
mary.

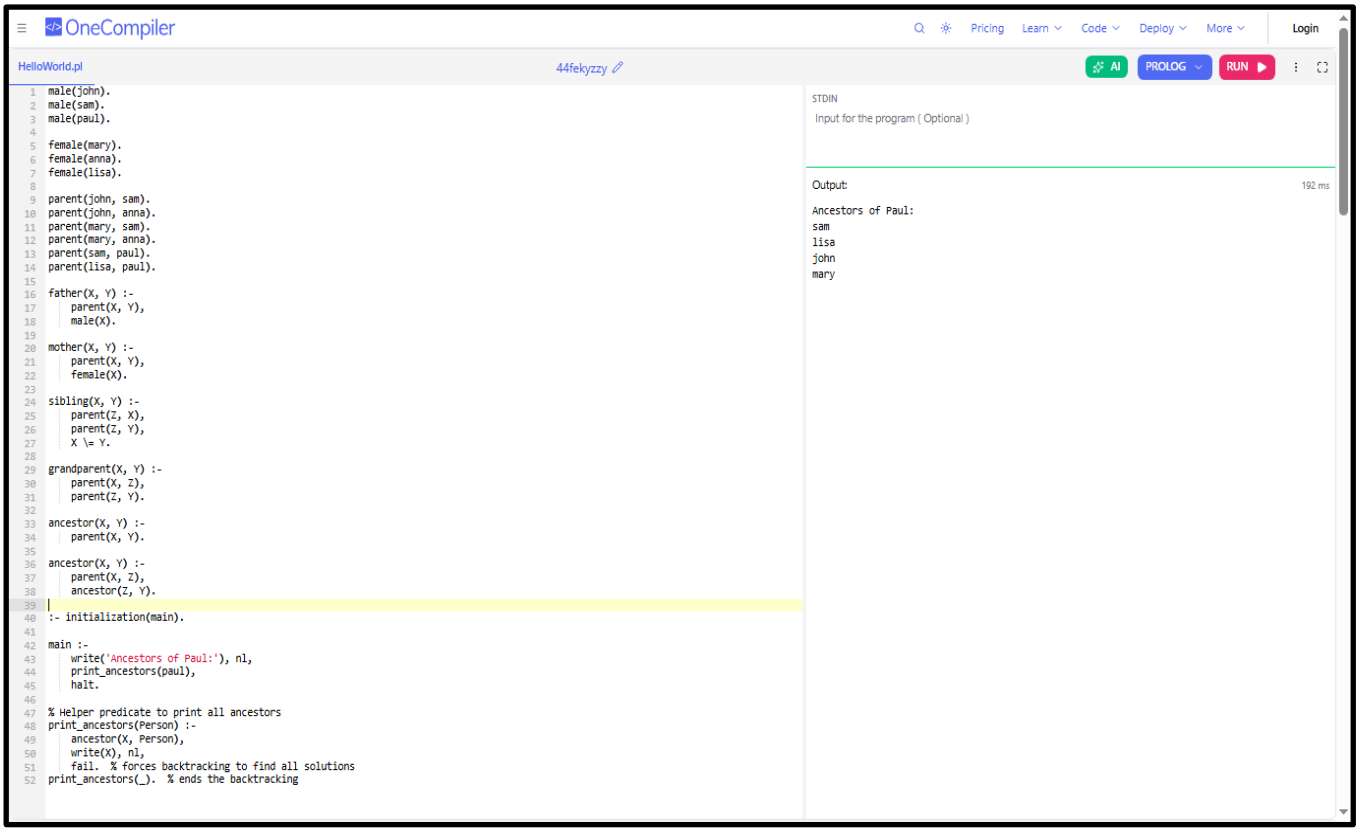
4. Find ancestors of Paul ?-

ancestor(X, paul).

Output:

X = sam ;
X = lisa ;
X = john ; X =
mary.

(Uses recursion + backtracking)




```
male(john).
male(sam).
male(paul).

female(mary).
female(anna).
female(lisa).

parent(john, sam).
parent(john, anna).
parent(mary, sam).
parent(mary, anna).
parent(sam, paul).
parent(lisa, paul).

father(X, Y) :-
    parent(X, Y),
    male(X).

mother(X, Y) :-
    parent(X, Y),
    female(X).

sibling(X, Y) :-
    parent(Z, X),
    parent(Z, Y),
    X \= Y.

grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).

ancestor(X, Y) :-
    parent(X, Y).

ancestor(X, Y) :-
    parent(X, Z),
    ancestor(Z, Y).

:- initialization(main).

main :-
    write('Ancestors of Paul:'), nl,
    print_ancestors(paul),
    halt.

% Helper predicate to print all ancestors
print_ancestors(Person) :-
    ancestor(X, Person),
    write(X), nl,
    fail.  % forces backtracking to find all solutions
print_ancestors(_).  % ends the backtracking
```

CONCLUSION: Hence we understood the concept of prolog also we Write a Prolog Program to demonstrate the concept of Family relationship.

DISCUSSION QUESTIONS:

1. What is a fact in Prolog?

_____

_____

_____

2. What is the difference between fact and rule?

3. What is unification?

4. Explain backtracking with example?

5. Why is recursion used in `ancestor` rule?

6. What is the difference between `=` and `is`?

REFERENCES:

- https://www.swi-prolog.org/
- https://www.tutorialspoint.com/prolog/prolog_introduction.htm
- https://www.geeksforgeeks.org/artificial-intelligence/prolog-an-introduction/
- https://www.youtube.com/watch?v=dKn-BbS_zQQ